Derivation appendix: 03 — scalar decays and LFV rates

VV couplings, loop-induced modes and LFV rates

1. h_0 is gauge-phobic exactly, and the VV couplings obey a sum rule

derived here — §2

Claim

$$g_{h_iVV} = g^{\mathrm{SM}}\,(R_S^{\mathsf{T}}\hat{n})_i = g^{\mathrm{SM}}(\cos\delta,\,0,\,\sin\delta), \qquad \sum_i g^2_{h_iVV} = (g^{\mathrm{SM}})^2$$

Result

$$R_S^{\mathsf{T}}\hat{n} = [\cos(\delta) \quad 0 \quad \sin(\delta)]$$

Independent checks

- R_A's first column verified to be the unit vacuum direction
- the SM reference read off the $\delta \to 0$ limit, not typed in
- W and Z couplings both equal $g^{\mathrm{SM}}\times$ the same projection
- the zero survives a generic vacuum angle φ_v, so it is orthogonality and not the $\sqrt{3}$ alignment

Would be falsified by: a non-zero middle projection at generic φ_v, or a $\delta \to 0$ reference differing from $ig_w m_W$

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2. The draft's Scenario-C relations hold exactly, for all δ

derived here — §4.1

Claim

$$Q_2(C) = Q_2(A), \quad Q_1(C) = \cos\delta\, Q_1(A) - \sin\delta\, Q_3(A), \quad Q_3(C) = \sin\delta\, Q_1(A) + \cos\delta\, Q_3(A)$$

Independent checks

- symbolic identity at general φ, θ_v, δ and symbolic G_k

Would be falsified by: a non-zero symbolic residual in any of the three relations

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3. lepton_mus(μ_3) inverts the notebook-02 §5 relations

derived here — §7

Claim

$$\mathrm{svd}\big(M_\ell(\mu_1,\ldots,\mu_4)\big) = (m_e,\,m_\mu,\,m_\tau) \qquad \text{for every } \mu_3 \in (m_\mu,\,m_\tau)$$

Independent checks

- singular values match the PDG masses to 10^{-9} relative across the whole μ_3 scan

- the Yukawas genuinely vary, so the check is not trivial
- a 1% perturbation of μ_4 breaks it

Would be falsified by: the spectrum matching at one μ_3 but not across the scan, which would mean the inversion is only locally right

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4. At the exact-S_3 vacuum $h_{1,2}$ give only $\tau\mu$, and h_0 only $e\tau$, $e\mu$

derived here — §7.2

Independent checks

- derived from the reflection structure of the doublet Yukawa blocks (section 7.2)
- SM-like BR(tau e), BR(e mu) and h_0 BR(tau mu) below 1e-20 at all 35514 entries

Would be falsified by: an SM-like electron coupling or an h_0 tau-mu coupling at any point

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5. The one-loop form factors $A_{1/2}$, A_1, A_0 and the LFV width formula

imported — §5–6 (from [Djouadi08])

Claim

$$\Gamma(h \to \ell_a \bar{\ell}_b + \bar{\ell}_a \ell_b) = \frac{m_h}{8\pi}(|Q_{ab}|^2 + |Q_{ba}|^2)$$

Independent checks

- $A_{1/2} \to 4/3$ and $A_1 \to -7$ in the decoupling limit
- $A_0 \to -1/3$, the standard charged-scalar value
- the SM diphoton width is reproduced against PDG

Would be falsified by: a decoupling limit disagreeing with the literature values

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